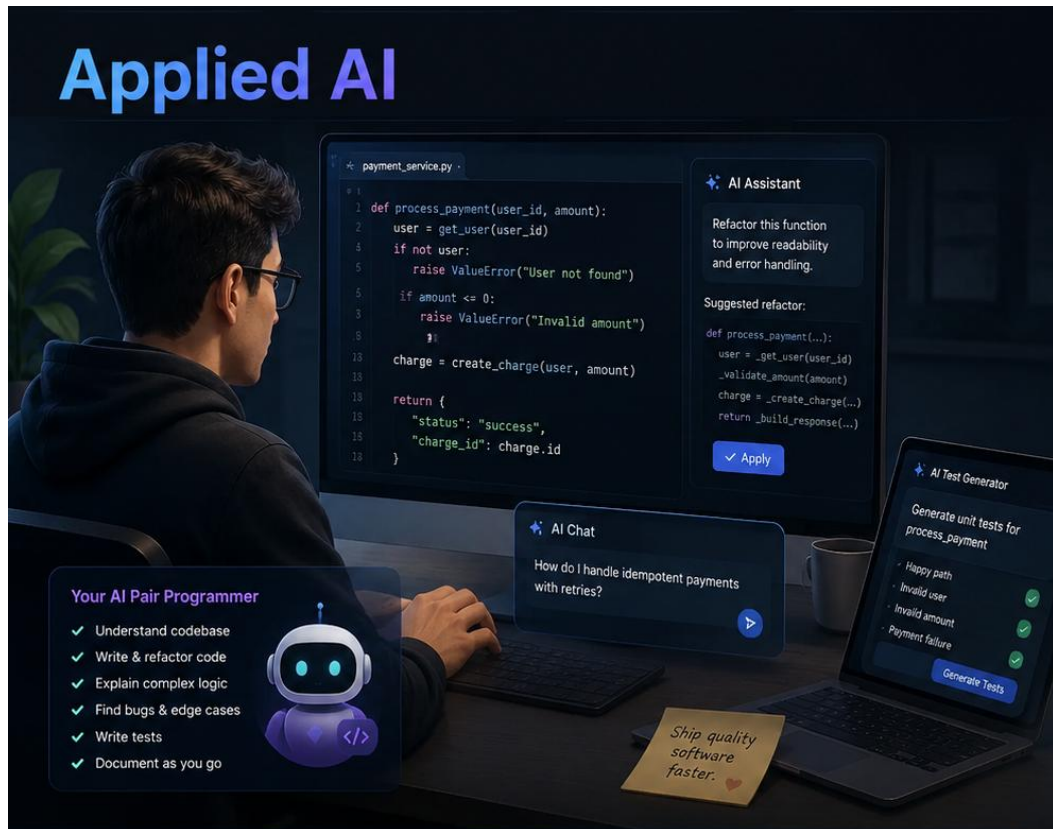


Applied AI

Module book

2026-2027
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Applied AI



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Preface

Artificial Intelligence (AI) is playing an increasingly important role in the business world and has become an essential technology. In the past, companies mainly relied on traditional software and manual processes. Today, AI enables organisations to analyse large amounts of data, make predictions, and automate repetitive tasks. As a result, business processes become more efficient, employees can focus on more complex work, and new opportunities for innovation emerge. Examples include AI-powered customer service chatbots, predictive maintenance in industry, and data-driven decision support systems.

For information technology students, this development means that the expectations of the job market have changed significantly. While programming skills alone were often sufficient in the past, today's professionals are also expected to understand how AI models work, how they can be applied responsibly, and how they can be integrated into existing systems. Applied AI is not only about developing algorithms but, more importantly, about using AI in practical ways to solve real business problems.

The difference compared to the past is clear. Previously, computers mainly functioned as tools that executed exactly what programmers instructed them to do. Modern AI systems, however, can learn from data, recognize patterns, and in some cases generate recommendations or support decision-making. As a result, the role of the computer scientist is evolving from that of a software developer to a professional who combines technology, data, and business processes.

Knowledge of applied AI is therefore no longer a specialised skill but a fundamental competency for future IT professionals. Students who learn how to apply AI effectively and responsibly will be better prepared for the labour market and will contribute to the digital transformation and innovation of organisations.

The study load for this module is 420 hours (15 EC) in the first period and again 420 hours (15 EC) in the second period.

ChatGPT & R. Loves
Emmen, 26-06-2026

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1 Introduction

Artificial Intelligence is increasingly becoming part of professional software development and wider IT practice. IT professionals are expected not only to understand the possibilities of AI, but also to determine when and how it can be applied effectively, responsibly and professionally.

The Applied AI module prepares students to work with AI in realistic professional contexts. Students combine technical knowledge with software development, experimentation, critical thinking and professional judgement. They learn to investigate real-world problems, explore appropriate AI-based solutions and evaluate both their technical value and their wider implications.

The module therefore places students in a professional setting in which they are expected to take increasing responsibility for their own work, learning and professional development.

Typical professional situation

In this module, students are placed in a professional context similar to that of a junior Applied AI engineer.

Alex has been working for a technology consultancy firm for one year, specializing in data-driven solutions. After completing a higher professional education programme in Computer Science with a focus on Applied Artificial Intelligence, Alex developed strong skills in machine learning, data analysis, and software development. During this programme, Alex also worked on practical AI projects involving real-world datasets and intelligent system design.

Within the organization, Alex contributes to projects that require the development and deployment of AI solutions. These projects range from predictive analytics to intelligent automation and decision-support systems.

Some of Alex's tasks include:

- Designing and implementing machine learning models for real-world problems.
- Collecting, preprocessing, and analysing data from various sources.
- Integrating AI components into existing software systems and workflows.
- Evaluating model performance and improving system reliability.
- Communicating results and insights to both technical and non-technical stakeholders.

In addition, Alex actively keeps up with the latest developments in artificial intelligence and data science. Continuous learning is essential, and the organization supports this through training opportunities, workshops, and professional certifications. Alex aims to further specialize in applied AI and contribute to innovative, responsible, and scalable AI solutions.

1.1 Competences

Within this module, students develop and demonstrate HBO-i competences related to analysing, managing and controlling, designing and realising IT solutions in the context of Applied AI.

	M&C	Analysis	Realisation
Software	> Manage and use a development environment to support software development in teams, including e.g. continuous integration (CI) as an option Apply methods and techniques to manage a software development process and ensure its quality (Level 2)	> Perform a requirements analysis for a software system with various stakeholders in a context of existing systems Define acceptance criteria based on quality characteristics and a performed risk analysis including sustainability, security and privacy aspects and accessibility (Level 3)	> Build, test and make available a scalable software system that ties in with existing systems, optionally in the cloud, according to a designed or generated architecture with substantiated use of frameworks > Draw up and perform regression, integration and system tests and evaluate, process and follow up on the results > Refactor an existing application using design patterns (Level 3)

1.2 Module theme

The central theme of this module is the practical application of Artificial Intelligence in professional IT contexts. Students learn how AI can be used to address real-world problems and how AI-enabled solutions can be analysed, designed, developed, evaluated and applied responsibly.

During the first four weeks, all students develop a common foundation in Applied AI. This includes the prescribed Path 0 online learning, guest lectures, workshops, self-study and the preparation of an academic research paper written in pairs. These activities provide a shared starting point before the project phase begins.

From week five onwards, students work in teams on an Applied AI project provided by an external organisation or an NHL Stenden research group. The project provides the professional context in which students apply and further develop their knowledge and skills. Although the project is carried out as a team, assessment takes place on an individual basis. Students must therefore be able to identify, explain and justify their own contribution to the project and demonstrate how this contribution provides evidence of the relevant learning outcomes and HBO-i competences.

The projects are approached in a structured and iterative manner, building on the Scrum methods students have already used earlier in the programme. Scrum provides a framework for organising work, setting priorities, dividing responsibilities and reviewing progress. Students are expected to move beyond preparation and to produce tangible results through investigation, experimentation, development, testing and evaluation. Feedback and emerging insights are used to determine subsequent steps and improve both the project approach and the work produced.

Throughout the project, students should be able to explain and justify why they are doing what they are doing. Choices concerning methods, technologies, tools, priorities, experiments and solutions should be deliberate, evidence-based and appropriate to the needs of the project.

Alongside the project, each student follows a project-aligned learning path. After completing the required foundation in the first period of the Applied AI module, students select an approved learning route that supports the knowledge and skills needed for their individual contribution to the project. This learning is complemented by relevant academic and professional literature. Students are expected to

demonstrate not only what they studied, but also how newly acquired knowledge and skills were applied within the project.

During the second period of the Applied AI module, each project group also produces a working paper based on the project. The working paper provides an opportunity to present relevant aspects of the project in an academic and evidence-based manner. Although the paper is produced by the group, each student must make a clear and identifiable individual contribution to its development.

Wednesday afternoons form the central contact moment of the module throughout the semester. Students spend approximately five hours each Wednesday participating in activities such as guest lectures, tutorials, workshops, project consultation, presentations, demonstrations and feedback sessions. The Wednesday sessions are also the primary opportunity to discuss questions concerning the module, project work, learning activities, expectations and assessment with the teaching team.

Professional development and responsible IT practice are integrated throughout the module. Students are expected to consider the ethical, societal and professional implications of Artificial Intelligence and to use the ACM Code of Ethics and Professional Conduct as an important professional framework. During the second period of the Applied AI module, students contribute to a conference in which the morning programme focuses on Ethics and AI and the afternoon programme focuses on presenting and explaining the technologies, techniques and approaches used within the Applied AI projects to students from earlier years.

Throughout the semester, students collect evidence of their individual work, learning, professional conduct and development in a portfolio. The portfolio is cumulative and forms the main evidence base for assessment. The module assessment consists of the go/no-go assessment at the end of the first period of the Applied AI module, the final portfolio assessment and the criterion-based interview at the end of the second period of the module.

1.3 Learning Outcomes

After completing this module, students can:

1. Analyse and select appropriate AI approaches

The student can analyse a professional problem, evaluate relevant AI techniques, models and technologies, and justify the selection of an appropriate approach based on the requirements, opportunities and limitations of the professional context.

2. Develop and evaluate AI-enabled solutions

The student can design, implement, integrate, test and evaluate AI-enabled software solutions using appropriate data, models, tools, frameworks and professional software development practices. The student can evaluate the quality and limitations of the resulting solution and use these findings to improve it.

3. Work in a structured and iterative project environment

The student can organise and carry out project work in a structured and iterative manner using Scrum. The student can translate project objectives into prioritised activities, contribute to planning and the division of responsibilities, produce tangible results in successive iterations, and use experimentation, evaluation and stakeholder feedback to determine appropriate next steps.

4. Direct and apply their own learning

The student can identify gaps in their own knowledge and skills, determine what additional learning is required, independently select and undertake appropriate learning activities, and apply newly acquired knowledge and skills within the project.

5. Apply AI responsibly

The student can assess the ethical, societal, legal and professional implications of an AI application, including relevant issues concerning data quality, bias, privacy, security, transparency, risks, regulations and standards, and translate these considerations into justified design and implementation choices.

6. Act as a responsible IT professional

The student can act professionally in the development and application of AI by collaborating and communicating effectively with technical and non-technical stakeholders, taking responsibility for their own work, giving and receiving professional feedback, explaining and justifying professional choices, critically reflecting on their own conduct, and considering the wider professional and societal responsibilities associated with computing and Artificial Intelligence.

1.4 Alignment of Learning Outcomes and Assessment

Learning outcomes 1 and 2 are primarily demonstrated through the student's individual project contribution. They provide evidence for HBO-i Analyse and Realise at level 3 and, where applicable, Design at level 3. Learning outcome 3 is demonstrated through Scrum and iterative project evidence and relates primarily to Manage & Control at level 2.

Learning outcome 4 is assessed through the individual learning path and its application within the project. Learning outcome 5 is demonstrated through responsible AI analysis and documented professional and technical decisions. Depending on the evidence, it supports Analyse, Design and Realise at level 3.

Learning outcome 6 is assessed through evidence of professional conduct, collaboration, communication and responsibility and relates primarily to Manage & Control at level 2.

Students identify and justify the strongest evidence for each learning outcome and applicable HBO-i competence in their portfolio. The criterion-based interview verifies the student's ownership, understanding and professional reasoning. The portfolio weighting and final pass requirements are specified in Appendix 8.

1.5 Prerequisite Knowledge

The prerequisite knowledge for taking the module has been included in the most recent version of the OER.

1.6 Conventions

The agreements that apply to the designation and structure of the program code can be found on Blackboard within ICT-general.

1.7 Version management

Version	Date	Author	Description
1.0	01-09-2026	Development team	Initial module book

1.8 Attendance and participation

Wednesday afternoons form the central contact moment of the Applied AI module. Students are expected to attend and actively participate throughout the semester.

The Wednesday programme may include guest lectures, workshops, project consultations, presentations, demonstrations, feedback sessions and other activities related to the projects and individual learning routes. Guest lecturers will regularly address specific aspects of AI in software development and provide students with perspectives from professional practice and research.

Attendance on Wednesday afternoons is compulsory. Active participation is important because many activities depend on interaction with lecturers, guest speakers, fellow students and project teams. Students are therefore expected to come prepared and to use these sessions to discuss progress, present results, ask questions, receive feedback and determine appropriate next steps.

The Wednesday sessions are also the primary opportunity for students to ask questions and obtain guidance from the teaching team. Questions concerning the module, project work, learning activities, expectations, assessment or examination should therefore be raised during these sessions wherever possible.

Students are expected to make active and effective use of the Wednesday afternoon sessions rather than relying on individual emails or Teams messages to lecturers for matters that can reasonably be discussed on Wednesday.

If a student is unable to attend for a valid reason, the student is responsible for informing the teaching team and catching up on the missed learning activities. An alternative assignment or other form of evidence may be required.

Additional compulsory activities, such as company visits, excursions or workshops, may also be scheduled during the semester on any available working day(s). Students will be informed about these activities in advance and are expected to attend.

2 Examination

2.1 Introduction

The assessment in this module reflects the applied and professional nature of Applied AI. Students are expected not only to demonstrate knowledge of Artificial Intelligence, but also to show that they can apply this knowledge effectively, responsibly and professionally within a realistic project context.

Students complete the Applied AI project in a team. Assessment, however, takes place entirely on an individual basis. The quality of the final team product is therefore not assessed as a separate result and does not determine an individual student's grade. Instead, the project provides the professional context and evidence through which each student demonstrates their own achievement.

Students are assessed on the basis of their individual contribution to the project, the quality and level of the work for which they were personally responsible, their understanding of that work, and their ability to explain and justify the choices they made. This includes their ability to analyse problems, investigate and compare possible approaches, make appropriate technical and design decisions, develop and evaluate solutions, work in a structured and iterative manner, and use feedback and evidence to improve subsequent work.

The individual project contribution is also an important source of evidence for demonstrating the relevant learning outcomes and HBO-i competences. Students must therefore be able to show not only what they contributed, but also how their contribution demonstrates the required level of analysis, realisation, project-based working and responsible application of AI. Where work has been produced jointly, the student must clearly identify their own role and contribution.

A strong final team product cannot compensate for insufficient individual achievement. Conversely, limitations in the overall team product do not automatically prevent an individual student from demonstrating the required level, provided that their own contribution, technical work, reasoning, evaluation and understanding provide sufficient evidence of the learning outcomes and competences.

Professional development is assessed as an integral part of the module. Students are expected to communicate and collaborate professionally, take responsibility for their work, critically reflect on their choices and actions, and demonstrate awareness of the

responsibilities associated with working as an IT professional in a field increasingly shaped by AI.

The following sections describe the evidence that students must collect in their portfolio and explain how this evidence is used to assess their individual contribution, professional development, learning outcomes and competences, and to determine successful completion of the module.

2.2 Portfolio

The portfolio is the central collection of evidence used throughout the Applied AI module. Students use the portfolio to collect, organise and submit evidence of their learning, development and contribution during the semester.

The portfolio is individual. Each student is responsible for ensuring that it provides a clear and complete overview of the work carried out, the learning activities undertaken and the results achieved. Evidence may include documents, project products, code, presentations, reflections, certificates, feedback, links to repositories and other relevant materials.

Students are expected to maintain the portfolio throughout the semester. It should therefore develop alongside the student's learning and project work rather than being assembled only shortly before an assessment.

The portfolio forms the basis for both the go/no-go assessment at the end of the first period of the Applied AI semester, and the final assessment at the end of the semester. The following sections specify the evidence that students are required to include for these assessments.

The portfolio must be submitted as a single, clearly structured PDF document. Where evidence cannot reasonably be included directly in the portfolio, such as source code, repositories, videos or online certificates, the portfolio must contain a clear reference or working link to that evidence.

For the required structure of the portfolio, please consult the relevant appendix to this module book.

2.3 Assessment after period 1: go/no-go

At the end of the first period of the Applied AI semester, each student takes part in an individual go/no-go assessment. The purpose of this assessment is to determine whether the student has

made sufficient progress during the first eight weeks to continue with the Applied AI module.

The assessment is based on the evidence collected in the student's portfolio. At this stage, the student must demonstrate that the required activities for period 1 have been completed and that sufficient progress has been made in both the project and the individual learning process.

To receive a go, the student must demonstrate that:

- the prescribed online courses have been completed successfully;
- An academic paper has been completed successfully;
- the student has received substantive peer feedback on the paper and has demonstrably considered this feedback;
- the student has provided substantive and constructive peer feedback to the paper of at least one other student;
- the student has made a demonstrable and meaningful contribution to the project since its start in week 5;
- sufficient evidence of progress has been collected in the portfolio;
- the student can explain and justify the work carried out, the choices made and the next steps in the project;
- the student has shown sufficient professional participation and responsibility during the first period of the module.

A go means that the student has demonstrated sufficient progress to continue with the project and the second period of the Applied AI module.

A no-go means that the student has not yet demonstrated sufficient progress or has not completed one or more of the required components. In that case, the student cannot continue with the project in the normal manner until the deficiencies have been addressed in accordance with the applicable assessment.

The go/no-go assessment is not the final assessment of the module. Its purpose is to verify that the student has established a sufficient foundation and has demonstrated enough progress to continue towards the final assessment at the end of the semester.

For further information about the requirements for the go/no-go assessment in the first period of the Applied AI module, students should consult the appendices to this module book. The prescribed structure and requirements of the portfolio are also included in the appendices.

2.4 Final assessment

The final assessment takes place at the end of the Applied AI semester and is conducted on an individual basis. The purpose of the assessment is to determine whether the student has achieved the learning outcomes and competences of the Applied AI module at the required level.

The assessment is based on the evidence collected in the student's portfolio and on the student's ability to explain, justify and reflect on this evidence. The portfolio must make clear what the student has personally contributed, what the student has learned during the semester, and how this learning has been applied in practice.

The portfolio used for the final assessment is cumulative. This means that all evidence submitted for the go/no-go assessment at the end of the first period of the module must also remain included in the portfolio submitted for the final assessment. The final portfolio therefore provides a complete overview of the student's learning, development and contribution throughout the entire semester.

As part of the final assessment, the student participates in an individual criterion-based interview with the assessors. During this interview, the student is questioned about the evidence presented in the portfolio and is expected to explain, justify and reflect on their work, choices, learning and professional development. The student can also be asked to present evidence of understanding in the form of live coding. The interview allows the assessors to establish the student's individual level of achievement and to verify that the submitted evidence genuinely reflects the student's own contribution and understanding.

The final assessment mainly focuses on three closely related areas:

- **Individual contribution to the project** – the student must demonstrate a substantial and identifiable contribution to the project and be able to explain and justify the methods, technical choices, activities and results for which the student was responsible.
- **Professional skills and conduct** – the student must demonstrate that they have worked and acted as a responsible IT professional. This includes collaboration, communication, professional responsibility, critical reflection and the ability to consider the wider professional and societal implications of the use of AI.
- **Individual learning route** – the student must demonstrate continued development of the knowledge and skills needed for their individual contribution to the project. After

completing Path 0 in period 1, the student follows a project-aligned learning path and studies relevant project literature. The portfolio must provide evidence of this learning and show how the acquired knowledge and skills have been applied within the project.

The final assessment therefore considers both what the student has produced and how the student has developed and acted throughout the semester. Students are expected to demonstrate ownership of their work and must be able to explain and justify why they made particular choices.

The requirements and evidence associated with each of these three areas are described in the following sections.

2.4.1 Individual contribution to the project

The Applied AI project is carried out in a team, but assessment takes place entirely on an individual basis. The project itself and the overall project result are not graded within this module. Instead, the project provides the professional context in which each student demonstrates their individual achievement of the relevant learning outcomes and HBO-i competences.

Each student must demonstrate a substantial, identifiable and relevant personal contribution to the project. The portfolio must contain sufficient evidence to allow the assessors to determine what the student personally did, which responsibilities they assumed, how their work developed and at what level they performed.

Evidence may include analyses, research, experiments, designs, architectural decisions, prototypes, source code, AI models or components, datasets and data-processing activities, tests, evaluations, technical documentation, presentations, feedback, Scrum artefacts, contributions to the project group's working paper and other relevant evidence. Where work has been produced jointly, the student must clearly identify and explain their own contribution.

The project group's working paper is a group deliverable, but each student must make a clear and substantive individual contribution to its development. This contribution may include, for example, literature research, methodology, analysis, technical content, interpretation of results, discussion or conclusions. The student must be able to identify which parts of the working paper they contributed to, explain the substance of that contribution, and show how it relates to their role and activities within the wider project.

The assessment considers both the technical and professional quality of the student's individual work and the reasoning underlying that

work. Students must demonstrate that they can analyse relevant problems and requirements, investigate and compare possible approaches, make substantiated technical and design choices, develop and evaluate solutions, and understand the limitations and consequences of their work. Their individual contribution must provide sufficient evidence of the applicable technical learning outcomes and HBO-i competences at the required level.

Students are also expected to demonstrate a structured and iterative way of working, building on the Scrum methods introduced earlier in the programme. They should be able to show how they planned and prioritised their work, produced tangible results, investigated uncertainties, experimented with possible solutions, evaluated outcomes, processed feedback and used emerging insights to determine subsequent steps.

Students must be able to explain and justify the choices for which they were responsible. This includes choices concerning methods, technologies, tools, architectures, experiments, priorities and solutions. They should be able to explain which alternatives were considered, why particular choices were made, how results were evaluated, what was learned and how these insights influenced subsequent work.

The student must also demonstrate understanding and ownership of their contribution. They must be able to explain not only what they produced, but also how their work relates to the wider project, which dependencies and limitations are involved, and why their contribution demonstrates the relevant learning outcomes and HBO-i competences. The same principle applies to the student's contribution to the working paper: the group paper itself does not determine the student's individual assessment, but the student's identifiable contribution to it may provide relevant evidence of their achievement.

Assessment is based on the evidence contained in the portfolio and the student's explanation, justification and reflection during the criterion-based interview. The performance of the project team or the quality of the overall project result cannot compensate for insufficient individual evidence. Conversely, limitations in the wider project do not prevent a student from demonstrating the required level if their own contribution provides sufficient evidence of achievement.

For the detailed assessment criteria, please consult the relevant appendices.

2.4.2 Professional skills and conduct

Students are expected to demonstrate that they have acted and developed as responsible IT professionals throughout the Applied AI module. Professional skills and conduct are therefore assessed on the basis of the student's behaviour, choices, professional judgement, communication and development throughout the semester rather than on a single activity.

The assessment considers how the student collaborates within the project team, communicates with fellow students, teachers, clients and other stakeholders, takes responsibility for agreed work, gives and receives feedback, responds to problems and contributes constructively to the progress of the project.

ACM Code of Ethics and Professional Conduct

An important framework for this part of the module is the ACM Code of Ethics and Professional Conduct. Students are expected to become familiar with the Code and to use relevant principles when reflecting on their own behaviour and on professional decisions encountered during the module. They should be able to recognise situations involving professional or ethical considerations, identify relevant stakeholders and interests, consider possible consequences and explain and justify their own professional choices.

Professional development is not limited to project work. Tutorials, guest lectures, workshops, discussions and other module activities are intended to broaden the student's understanding of the responsibilities associated with working as an IT professional in a field increasingly influenced by Artificial Intelligence. Students are expected to make active use of these activities and to reflect on how they contribute to their professional development. In this reflection students are expected to show knowledge and understanding of the ACM Code of Ethics and Professional Conduct.

Ethics & AI Conference

As part of this professional development, an Ethics & AI Conference is organised during period 2. The conference provides students with an opportunity to share what they have learned with students from earlier years of the Information Technology and Computer Science programmes.

The morning programme focuses on Ethics and AI. Applied AI students prepare and deliver lectures, workshops or other suitable educational activities concerning ethical, societal and professional issues associated with Artificial Intelligence. Students are expected to demonstrate sufficient understanding of their chosen subject, communicate it clearly to the intended audience, encourage critical thinking and discussion, and respond professionally to questions and

different viewpoints. The ACM Code of Ethics and Professional Conduct provides an important point of reference for these activities.

The afternoon programme focuses on the Applied AI projects. Each project group presents its work through a poster presentation and contributes workshops or other educational activities in which students explain relevant technologies, techniques, methods or approaches used within their project. Every member of the project group is expected to make an active and identifiable contribution. The aim is not only to present project results, but also to help first-, second- and, where applicable, third-year students understand and engage with the technologies and techniques being used.

Students should therefore be able to adapt their communication to the knowledge and experience of their audience, explain complex technical and professional topics clearly, answer questions, respond constructively to different viewpoints and contribute to the learning of others.

The organisation and practical coordination of the conference are the responsibility of the teaching team and supporting staff. Applied AI students are responsible for the preparation, quality and delivery of their own contributions.

Assessment of Professional Skills and Conduct

The portfolio must contain concrete evidence of the student's professional skills, conduct and development. Evidence should originate from different parts of the semester, including project work and the conference. General statements about professional behaviour or development are not sufficient; claims must be supported by concrete situations, reflection and relevant evidence.

During the criterion-based interview, students must be able to explain and reflect on their professional conduct and development. They should be able to discuss situations in which professional responsibilities, ethical considerations, communication challenges or different stakeholder interests had to be weighed and explain why they acted, or would act, in a particular way.

The required portfolio evidence for this assessment component is described in the relevant appendix. Professional skills and conduct form part of the portfolio assessment and are assessed using the applicable assessment criteria in the final assessment form.

2.4.3 Individual learning route

In addition to the common foundation provided during period 1, each student is expected to follow an individual learning route throughout

the project. The purpose of this learning route is to develop the knowledge and skills the student needs to contribute effectively to the project and to support their further development as an Applied AI professional.

From the start of the project, students are expected to identify gaps in their own knowledge and skills and determine what additional learning is required. These learning needs will differ between students depending on the project, their role within the team and the tasks for which they take responsibility.

Students are expected to select appropriate learning activities themselves. These may include online courses, tutorials, workshops, professional literature, technical documentation, experimentation or other relevant learning resources. The choice of learning activity must be appropriate to the identified learning need and should be justified by the student.

The portfolio must provide evidence of the individual learning route. For each significant learning activity, the student should demonstrate:

- what knowledge or skill needed to be developed;
- why this learning need was relevant to the project and the student's role;
- which learning activity was undertaken and why it was selected;
- evidence that the learning activity was completed;
- how the acquired knowledge or skills were applied within the project;
- what the student learned and how this influenced subsequent work.

Completion of a course or other learning activity alone is not sufficient. Students must demonstrate that the learning contributed to their professional development and, where relevant, was applied in the project.

During the criterion-based interview, students must be able to explain and justify the choices made within their individual learning route and reflect on how their knowledge and skills developed during the semester.

The individual learning route is assessed using the applicable scoring rubric included in the appendices to this module book.

2.4.4 Additional assessment components

In addition to the assessment of the student's individual contribution to the project, professional skills and conduct, and individual learning route, the final assessment takes into account several additional

components. Together, these provide the assessors with a broader view of the student's achievement and development throughout the semester.

External client feedback

The external client is asked to provide feedback on the student's performance during the project. This feedback is structured around the Dublin Descriptors and considers the student's ability to apply knowledge and understanding in a professional context, make informed judgements, communicate effectively, and demonstrate the capacity for further independent learning.

The client's feedback provides an important external perspective on the student's performance in a realistic professional setting. It forms part of the evidence considered by the examiners but does not independently determine the final assessment.

The completed client feedback must be included in the portfolio.

Learning outcomes and competence justification

Students must demonstrate how the evidence contained in their portfolio supports achievement of the learning outcomes and HBO-i competences specified for the Applied AI module.

For each learning outcome and competence, the student identifies the strongest and most relevant evidence and briefly explains why this evidence demonstrates achievement at the required level. Where evidence concerns work produced within a team, the student must clearly identify their own contribution.

The required format for the learning outcomes and competence justification is included in the appendices to this module book.

Overall reflection

The portfolio must contain an overall reflection on the student's development throughout the Applied AI semester. This reflection should bring together the different parts of the student's learning rather than discuss them as separate activities.

The student should reflect on important learning experiences, feedback received, strengths and areas for further development, and the way in which the semester has contributed to their development as an IT professional. The reflection should also consider how the student's understanding of Applied AI and its role within professional IT practice has developed.

The reflection should be critical and supported by concrete experiences and evidence. A primarily descriptive account of activities undertaken during the semester is not sufficient.

Quality and readability of the portfolio

The portfolio itself is expected to meet an appropriate professional standard. It must be complete, clearly structured and easy for the assessors to navigate.

The quality of the portfolio is considered in terms of its organisation, readability, consistency and the clarity with which evidence is presented and referenced. Links and references to supporting evidence must be clear and accessible.

The portfolio should enable the assessors to understand what the student has done, what the student has learned, how the student has developed and how the submitted evidence supports the claims made. The emphasis is on clear and effective communication rather than elaborate presentation.

These additional assessment components form part of the evidence discussed during the criterion-based interview and are assessed using the applicable scoring rubric included in the appendices to this module book.

2.5 Module resits

Students who do not meet the requirements of the Go/no-go assessment at the end of period 1 are given the opportunity to remedy the identified deficiencies. Because the project continues during period 2, this resit must take place within a short period of time.

The student must submit an improved version of the portfolio no later than the end of week 2 of period 2. The improved portfolio must address the deficiencies identified during the first assessment. Based on this resubmission, the examiners determine whether the student has demonstrated sufficient progress to continue with the module.

Students who do not successfully complete the final assessment are also entitled to a resit. The resit may take place in the weeks following the first final assessment. The exact date, form and required improvements are determined in consultation between the student and the examiners, based on the deficiencies identified during the first assessment.

When an improved portfolio is required for the final resit, it must be submitted no later than eight weeks after the date of the first final assessment.

The resit focuses on the assessment components that were found to be insufficient. Students are expected to use the feedback from the first assessment to improve the relevant evidence and to demonstrate that the required learning outcomes and competences have subsequently been achieved.

3 Programme

3.1 Introduction

The Applied AI module combines structured learning activities, professional input and project-based learning. During the first four weeks, students develop a common foundation in Applied AI through prescribed online learning, lectures, workshops and the paper assignment. From week 5 onwards, the external project becomes the central learning environment of the module.

Throughout the semester, Wednesday afternoons form the fixed contact moment of the programme. These afternoons include guest lectures on specific aspects of AI in software development and may also be used for project guidance, consultation, presentations, demonstrations, workshops and feedback sessions.

During the project phase, students work in teams and develop their solutions iteratively. At the same time, each student follows an individual learning route based on the knowledge and skills required to contribute effectively to the project.

This chapter describes the teaching methods, overall programme structure and weekly activities of the Applied AI module.

3.2 Teaching methods

The Applied AI module is based on the principle that students are responsible for directing their own learning. Teachers and experts provide guidance, expertise and support, but students are expected to identify what they need in order to progress in their project and individual learning route.

A broad range of expertise is available within the teaching team, the research groups and the wider professional network surrounding the module. Students are expected to make active use of this expertise. If they require additional explanation, technical guidance, a lecture on a particular subject, a workshop, access to specific expertise or another form of support, they are expected to identify this need and discuss it with the teaching team. Where possible and relevant, appropriate support will then be organised.

The teaching methods used within the module therefore respond partly to the needs that emerge during the semester. They include tutorials, guest lectures and workshops.

Tutorials

Tutorials are interactive and largely student-driven sessions. Students are expected to prepare through self-study and project work and to bring forward the questions, problems and topics for which they require guidance.

The teacher supports students in analysing problems, considering possible approaches and determining appropriate next steps. The purpose of a tutorial is not to provide all required knowledge in advance. Instead, it supports students in taking responsibility for their own learning and making informed decisions about how to proceed.

Tutorials also provide an important opportunity for students to discuss what they need in order to progress. Students are encouraged to raise questions about the programme, expectations, assessment and examination, and to seek clarification where necessary.

Tutorials may also be used for project consultation, discussion of intermediate results, feedback, technical guidance or the exploration of new learning needs.

Guest lectures

Guest lectures take place throughout the semester, primarily on Wednesday afternoons. Guest lecturers from professional practice, research and other relevant fields address specific aspects of AI in software development.

The guest lectures expose students to current developments, technologies, applications and professional considerations. Students are expected to participate actively, ask questions and consider how the topics discussed relate to their project, individual learning route and development as an IT professional.

Workshops

Workshops focus on the practical development and application of knowledge and skills. Students work actively with technologies, methods, tools or professional issues related to Applied AI.

Workshops may be scheduled as part of the programme, but may also arise from learning needs identified by students or project teams. Where several students encounter a similar knowledge or skills gap, the teaching team may organise a workshop or invite an appropriate expert to address that need.

The emphasis is on active participation, experimentation and learning by doing.

3.3 Curriculum Overview

The overview below shows the intended rhythm and main activities of the Applied AI semester. It is deliberately presented at a general level. Applied AI is project-driven and students are expected to take responsibility for planning their work, identifying learning needs and determining appropriate activities and next steps in consultation with their project team, client and teaching team.

During the first four weeks, the main emphasis is on establishing a common Applied AI foundation through the research paper and prescribed Path 0 learning. From week 5 onwards, the project becomes the central learning environment. Students increasingly combine project work with their individual learning path, literature study, professional development and portfolio development.

During period 2, project work continues in an iterative manner. Depending on the project, students may analyse problems and requirements, investigate alternatives, experiment, develop and integrate solutions, test and evaluate results, process feedback and refine their work. These activities do not necessarily take place in a fixed sequence: their timing and emphasis depend on the project and the decisions made by the students and project teams.

The Applied AI Conference provides an important shared activity during period 2. Towards the end of the semester, increasing attention is given to completing the project and working paper, consolidating portfolio evidence, reflection and preparation for the final assessment.

The overview is indicative and may be adjusted during the semester where required by project developments, student learning needs, availability of guest lecturers or other practical considerations. Changes will be communicated to students in a timely manner.

Students are responsible for keeping their portfolio up to date throughout the entire semester. Evidence, reflections and other relevant material should be added as the module progresses. It is not realistic to assemble or bring the complete portfolio up to the required standard shortly before submission deadlines.

Week	Task. Nr.	Study Activity
1.1	3.3.1	Start paper Applied AI Working on prescribed online modules

1.2	3.3.2	Consultation hours / working on the paper Working on prescribed online modules
1.3	3.3.3	Peer feedback papers Working on prescribed online modules
1.4	3.3.4	Hand in the paper for Go/no-go Working on prescribed online modules
1.5	3.3.5	Start project phase Working in project groups Working on prescribed online modules
1.6	3.3.6	Working in project groups Working on prescribed online modules
1.7	3.3.7	Working in project groups Working on prescribed online modules
1.8	3.3.8	Working in project groups Working on prescribed online modules End of week: Hand in portfolio for go/no-go assessment
T1	3.3.9	Go/no-go assessments Working in project groups Working on individual learning path
2.1	3.3.10	Working in project groups Working on individual learning path Building portfolio
2.2	3.3.11	Working in project groups Working on individual learning path Building portfolio Preparing Ethics and AI Conference
2.3	3.3.12	Working in project groups Working on individual learning path Building portfolio Preparing Ethics and AI Conference
2.4	3.3.13	Working in project groups Working on individual learning path Building portfolio Preparing Ethics and AI Conference
2.5	3.3.14	Working in project groups Working on individual learning path Building portfolio Ethics and AI Conference
2.6	3.3.15	Working in project groups Working on individual learning path Building portfolio

2.7	3.3.16	Working in project groups Working on individual learning path Building portfolio
2.8	3.3.17	Working on individual learning path Working on individual learning path Finishing portfolio Deadline for delivering the portfolio for final assessment
T2a		No final assessment
T2b	3.3.18	Final assessment / Criterion-based Interviews

4 Structure & Organisation

4.1 Structure & Organisation

Applied AI is a 30 EC module with a total study load of 840 hours. The module consists of two periods of 15 EC, corresponding to 420 study load hours per period.

Each period contains eight regular teaching weeks. During these weeks, Wednesday is the fixed contact day for Applied AI. Students are expected to spend each Wednesday attending scheduled module activities.

The Wednesday programme may include guest lectures, tutorials, workshops, project consultation, presentations, demonstrations, feedback sessions and other activities related to the module. Outside the Wednesday sessions, students organise the remainder of their study load independently.

Depending on the stage of the module and the needs of the project, this may include:

- online learning;
- paper preparation and writing;
- self-study and literature study;
- project work and collaboration;
- individual learning activities;
- meetings with clients, lecturers or other experts;
- preparation and collection of portfolio evidence;
- preparation for assessment.

Students are responsible for planning their study time appropriately. The exact distribution of hours between these activities will vary between students and projects.

4.2 Period 1

Period 1 consists of eight regular teaching weeks, followed by one test week. During the eight regular weeks, students attend the Wednesday sessions and work on the prescribed online learning, the period 1 paper, self-study and, from the start of the project phase onwards, project activities. The Go/no-go assessment takes place during the test week at the end of period 1.

4.3 Period 2

Period 2 also consists of eight regular teaching weeks. During these weeks, the project forms the main learning environment. Students combine project work with their project-aligned learning path, professional development activities, paper writing and preparation of the final portfolio.

The eight regular weeks are followed by two test weeks: T2a and T2b. The final assessment of Applied AI takes place during T2b. The criterion-based interview is also conducted during this test week. No final assessment is scheduled during T2a.

4.4 Overview of study load

Period	Regular teaching weeks	Scheduled Wednesday sessions	Assessment period	Study load
Period 1	8	8 × 5 hours	1 test week – Go/no-go assessment	420 hours
Period 2	8	8 × 5 hours	T2a and T2b – final assessment in T2b	420 hours
Total	16	80 hours	3 test weeks	840 hours

The scheduled Wednesday sessions account for approximately 80 hours across the semester. The remaining study load is devoted to the other learning, project and assessment activities described above.

5 Learning resources and software

The Applied AI module does not prescribe a fixed body of literature for the entire semester. The sources and learning materials used by students depend on the topic of their papers, the nature of the external project and the individual learning needs identified during the student's personal learning route.

Students are expected to select appropriate and reliable learning resources themselves and to be able to explain why particular sources, courses, documentation or other materials are relevant to their work.

A limited number of learning resources are prescribed for all students. These provide a common foundation in Applied AI and professional conduct and are described below.

5.1 Prescribed learning resources

Prescribed online modules

During period 1, all students complete the prescribed Path 0 online modules. These modules provide a common foundation in Applied AI and prepare students for the project phase of the module.

During period 2, students continue their online learning through a project-aligned learning route. Students select at least one route from the menu described in the appendices, may combine elements from different routes, or may propose a comparable alternative with approval from the teaching team. The selected learning activities should support the knowledge and skills needed for the student's individual contribution to the project.

The prescribed modules, available learning routes and requirements for completion or progress are described in the appendices to this module book.

Evidence of the student's online learning and its application within the project must be included in the portfolio.

ACM Code of Ethics and Professional Conduct

The [ACM Code of Ethics and Professional Conduct](#) is a prescribed professional framework throughout the Applied AI module.

Students are expected to familiarise themselves with the Code and to use it when considering their responsibilities as IT-professionals. The Code is relevant not only to explicitly ethical questions, but also to everyday professional practice, including collaboration, quality of work, honesty, professional competence, privacy, responsibility towards stakeholders, professional review and the consideration of possible consequences of computing systems.

Students are expected to apply and reflect on relevant principles from the ACM Code during their project work and other activities within the module. They should be able to relate concrete professional situations and decisions to the Code and explain how its principles have influenced, or could have influenced, their actions.

The ACM Code also provides an important framework for the professional and ethical discussions that form part of the Applied AI conference. Students are expected to consider how the development and application of AI affect the responsibilities of IT-professionals and to communicate these issues to others.

Evidence of this professional development and reflection must be included in the student's portfolio and forms part of the assessment of professional skills and conduct.

5.2 Student-selected learning resources

Beyond the prescribed and chosen learning paths, students are responsible for selecting the learning resources needed for their paper, project work and individual learning route.

Depending on the learning need, these resources may include:

- scientific and professional literature;
- books and articles;
- technical documentation;
- online courses and tutorials;
- documentation provided by software vendors or AI model providers;
- lectures and workshops;
- relevant professional standards and frameworks;
- other reliable learning resources.

Students are expected to critically evaluate the quality, relevance and reliability of the resources they use.

The selection of learning resources forms part of the student's individual learning process. Students should be able to explain what they needed to learn, which resources they selected and how the acquired knowledge or skills were subsequently applied.

5.3 Software and tools

No single software environment or fixed set of AI tools is prescribed for all projects. The technologies used will depend on the requirements of the project and the choices made by the project team.

Students are expected to select appropriate tools, technologies, frameworks and development environments and to be able to justify these choices.

If students require additional resources in order to carry out the project, such as software licences, cloud services, hardware, datasets or access to specific tools or environments, they must first discuss this need with the client. Where additional support or resources are still required, the matter should subsequently be discussed with one of the teachers involved in the Applied AI module.

Where appropriate, the teaching team can help students identify suitable expertise, facilities or alternative solutions. However, students remain responsible for identifying their needs in time and for taking the initiative to discuss them.

Students are also responsible for ensuring that the software, services and data they use comply with applicable requirements concerning privacy, security, licensing, intellectual property and responsible use of AI. In doing so, students are expected to consider relevant principles from the ACM Code of Ethics and Professional Conduct, particularly where their choices may affect users, stakeholders, data protection, system quality or professional responsibility.

5.4 Project expenses and travel

If students expect to incur costs in the course of their project, they are expected to discuss this in advance with the teaching team. This applies in particular to travel costs, but may also apply to the purchase of materials, software, services or other resources required for the project.

Possible reimbursement of expenses must be agreed with the teaching team beforehand. Students should therefore not assume that costs incurred as part of the project will automatically be reimbursed.

Students must retain invoices, receipts and other relevant proof of payment for all purchases and expenses related to the implementation of the project. Where reimbursement has been agreed, this documentation will be required in order to process the claim.

6 Module evaluation

The Applied AI module will be evaluated by an external evaluator who is not part of the Information Technology programme nor one of the research groups involved in the Applied AI course. The purpose of this independent evaluation is to assess the organisation, content, teaching methods, assessment approach and overall quality of the module.

The evaluation will be conducted through an interview with the students participating in the module. Based on this interview, the external evaluator will prepare a written evaluation report containing the main findings, conclusions and recommendations.

The report will be shared with the students, the teaching team, the Programme Committee and the management of the Information Technology programme. The report will serve as the basis for the further evaluation and improvement of the Applied AI module for future academic years.

7 Appendices

Appendix 1	Portfolio structure overview
Appendix 2	Learning path
Appendix 3	Research paper assignment
Appendix 4	Working paper assignment
Appendix 5	Professional skills portfolio
Appendix 6	External client feedback form (file in Teams environment)
Appendix 7	Criterion-based interview
Appendix 8	Grade calculation
Appendix 9	Assessment research paper (file in Teams environment)
Appendix 10	Assessment working paper (file in Teams environment)

Appendix 1:

Portfolio Structure Overview

The portfolio is an individual assessment document and must be submitted as a single, clearly structured PDF document. The portfolio may contain a maximum of 50 pages, excluding the paper and, where applicable, appendices specifically permitted by the teaching team.

Students are expected to select evidence for its relevance and quality rather than its quantity. The portfolio should provide the assessors with a clear and convincing overview of the student's learning, development and individual contribution throughout the semester.

The portfolio is developed in two stages. For the go/no-go assessment at the end of period 1, students are only required to submit the part of the portfolio prescribed for the period 1 assessment. The remaining sections do not yet need to be completed at this stage.

For the final assessment at the end of the semester, the complete portfolio must be submitted and all sections described in this appendix must be included. The final portfolio is cumulative: all evidence submitted for the go / no-go assessment must remain part of the portfolio and is supplemented with the evidence collected during period 2.

The portfolio must contain the following sections:

Section 1. Introduction and student information

A title page containing:

- the student's name and student number;
- the project title;
- the client or research group;
- the project team members;
- data and version;
- the student's main role or responsibilities within the project.

Section 2. Period 1 – Common foundation

Evidence of the activities and progress required for the go/no-go assessment at the end of period 1. This includes:

- evidence of successful completion of the prescribed online modules;
- the completed research paper;
- evidence of substantive peer feedback received on the paper;
- evidence of substantive and constructive peer feedback provided to at least one other student;
- where relevant, evidence showing how received feedback was processed;
- evidence of sufficient progress and individual effort during the first weeks of the project, clearly demonstrating what the student personally contributed after the start of the project in week 5.

This section must be present in both the portfolio submitted for the go / no-go assessment and the final portfolio.

Section 3. Project and project approach

A concise description of the project, including:

- the project context;
- the client or research group;
- the problem or challenge being addressed;
- the intended result;
- the student's role within the team;
- the way in which Scrum was used to structure the project;
- the way in which the project developed through successive iterations.

This section should provide sufficient context for the assessors to understand the evidence presented elsewhere in the portfolio. It is not intended to become a separate project report.

Section 4. Individual contribution to the project

This chapter contains evidence demonstrating the student's own contribution to the project. Relevant evidence may include analyses, experiments, designs, prototypes, source code, AI models or components, tests, research, technical documentation, presentations, feedback, Scrum artefacts and other relevant project results.

For significant contributions, the student should make clear:

- what the student personally did;

- why this work was undertaken;
- which approach was selected and why;
- what the result was;
- how the result was evaluated;
- what was learned;
- how the findings influenced subsequent work or iterations.

Where evidence concerns work produced by several team members, the student must clearly identify their own contribution. The detailed assessment criteria are included in the Individual Contribution to Project Scoring Rubric.

Section 5. Project group working paper

During the second period of the module, each project group produces a working paper related to the Applied AI project. The purpose of the working paper is to present relevant aspects of the project in an academic and evidence-based manner and to demonstrate appropriate use of literature, analysis, argumentation and reporting.

The working paper is a group deliverable. Each student must nevertheless make a clear and identifiable individual contribution to its development. Contributions may include literature research, analysis, methodology, technical content, interpretation of results, discussion, conclusions or other substantive parts of the paper.

The working paper must be included in, or clearly referenced from, the appendices of the final portfolio. Each student must briefly indicate:

- which parts of the working paper they contributed to;
- what their substantive contribution consisted of;
- how this contribution relates to their role and activities within the project.

The working paper is assessed using the Paper Scoring Rubric included in the appendices and or the Applied AI Teams environment. The student's identifiable individual contribution to the working paper also forms part of the evidence used in assessing their individual contribution to the project.

Section 6. Individual learning route

This section contains evidence of the student's project-aligned learning undertaken after completion of Path 0. Each student must follow at least one of the learning routes described in Appendix 2,

combine elements from different routes, or follow an alternative route approved by the teaching team.

The portfolio should make clear:

- which learning route or combination of learning activities was followed;
- why the selected learning was relevant to the student's role and contribution to the project;
- evidence of completion;
- student-authored work, such as adapted examples, exercises or other applications of the learning;
- the main learning highlights;
- how acquired knowledge or skills were applied within the project;
- a short reflection on what was learned and how this influenced subsequent work.

The section must also include evidence of the required project literature study. Each student must screen at least ten relevant abstracts and read at least five papers in depth, or a higher number where this is required by the project. The student should make clear how the literature contributed to their understanding of the subject and supported their work within the project.

Detailed evidence of the literature study, such as the list of screened abstracts, papers read and supporting notes, may only be included in the appendices of the portfolio. This material does not count towards the maximum of 50 pages for the main portfolio.

Students may use tools such as NotebookLM and other AI tools to support their learning and literature study. Where relevant, links to supporting notebooks or interactions may be included as evidence. Official certificates from online learning providers are not required. The emphasis is on demonstrating meaningful learning and its application within the project.

Section 7. Professional skills and conduct

This section contains the evidence required under the Professional Skills Portfolio Requirements and consists of two main components.

Professional Conduct and Ethics Reflection

The student must analyse at least four significant professional situations encountered during the semester. These situations are considered in relation to relevant principles from the ACM Code of Ethics and Professional Conduct.

At least one situation must relate directly to an ethical, societal or professional issue arising from the development or use of Artificial Intelligence.

Ethics and AI Conference

The student must provide evidence of their individual contribution to the Applied AI Conference, which is attended by first- and second-year Information Technology and Computer Science students. This includes evidence of their contribution to the Ethics and AI programme in the morning and to the project-focused programme in the afternoon.

The portfolio should demonstrate the student's substantive and educational contribution, including their role in preparing and delivering conference activities and what they learned from explaining and discussing relevant topics with others.

The complete evidence requirements are described in the relevant appendix.

Section 8. External client feedback

The completed external client feedback form must be included in the portfolio. The feedback provides an external perspective on the student's performance within the professional project environment and is structured around the Dublin Descriptors. The client feedback forms part of the evidence considered by the examiners but does not independently determine the final assessment. The client feedback form can be found in the Teams environment of the Applied AI module.

Section 9. Learning outcomes and competence justification

The student must provide an overview showing how the evidence contained in the portfolio demonstrates achievement of the learning outcomes and HBO-i competences of the Applied AI module. Both learning outcomes and HBO-i competences are listed in chapter 1 of this module book.

For each learning outcome and competence, the student should:

- identify the strongest and most relevant evidence;
- clearly indicate where this evidence can be found in the portfolio;
- briefly explain why the evidence demonstrates achievement;
- where relevant, identify the student's individual contribution;
- for HBO-i competences, justify why the evidence demonstrates achievement at the required level.

The purpose of this section is not to repeat evidence already contained elsewhere in the portfolio, but to make the relationship between the submitted evidence, learning outcomes and competences clear to the assessors.

Section 10. Overall reflection

The portfolio must contain a concise but critical overall reflection on the student's learning and development throughout the Applied AI semester.

The reflection should bring together the student's:

- technical development;
- individual learning;
- project experiences;
- professional development;
- feedback received;
- strengths and areas for further development;
- development as an IT professional.

The ACM Code of Ethics and Professional Conduct must form an explicit point of reference. The student should reflect on how their understanding of professional responsibility has developed, which ACM principles have become particularly relevant, and how experiences during the project and the Ethics and AI Conference have influenced their understanding of ethical and professional conduct.

The reflection should connect experiences from different parts of the semester rather than simply provide a chronological description of activities undertaken.

Section 11. References and external evidence

Where evidence cannot reasonably be included directly in the PDF, such as source-code repositories, demonstrations, videos, online certificates or other digital project products, the portfolio must contain a clear reference or working link. The student is responsible for ensuring that all referenced evidence is accessible to the assessors at the time of assessment.

Appendix 2:

Learning path

All students complete Path 0 during the first four weeks. The total workload is approximately 20 hours, planned for approximately five hours per week. This includes the course content, reproducing or adapting examples, recording learning highlights, and relating the learning to the module.

Path 0: required foundation

- **Agentic AI** <https://www.deeplearning.ai/courses/agent-ai>
- **AI Prompting for Everyone** <https://www.deeplearning.ai/courses/ai-prompting-for-everyone>

After Path 0, each student selects at least one project-aligned learning route. Students may combine courses or propose another resource, subject to approval by the teaching team.

Path 1: AI coding, code review and RAG

- **AI Coding Workflows: From Cloud to Local** <https://www.deeplearning.ai/courses/ai-coding-workflows-from-cloud-to-local>
- **Spec-Driven Development with Coding Agents** <https://www.deeplearning.ai/courses/spec-driven-development-with-coding-agents>
- **AI Code Review** <https://www.deeplearning.ai/courses/ai-code-review>
- **Retrieval Augmented Generation (RAG)** <https://www.deeplearning.ai/courses/retrieval-augmented-generation>

Path 2: LangChain and LangGraph

- **AI Agents in LangGraph** <https://www.deeplearning.ai/courses/ai-agents-in-langgraph>
- **Long-Term Agentic Memory** <https://www.deeplearning.ai/courses/long-term-agentic-memory-with-langgraph>
- **Functions, Tools and Agents** <https://www.deeplearning.ai/courses/functions-tools-agents-langchain>

- **Chat with Your Data**
<https://www.deeplearning.ai/courses/langchain-chat-with-your-data>
- **LangChain for LLM Application Development**
<https://www.deeplearning.ai/courses/langchain>

Path 3: Hugging Face

- **Open Source Models with Hugging Face**
<https://www.deeplearning.ai/courses/open-source-models-hugging-face>
- **Quantization Fundamentals**
<https://www.deeplearning.ai/courses/quantization-fundamentals>
- **Code Agents with smolagents**
<https://www.deeplearning.ai/courses/building-code-agents-with-hugging-face-smolagents>
- **Machine Learning Demos with Gradio**
<https://www.deeplearning.ai/courses/huggingface-gradio>

Path 4: Anthropic and Claude

Choose only when access does not require personal payment.
Otherwise, agree a no-cost alternative.

- **MCP with Anthropic**
<https://www.deeplearning.ai/courses/mcp-build-rich-context-ai-apps-with-anthropic>
- **Claude Code** <https://www.deeplearning.ai/courses/claude-code-a-highly-agentic-coding-assistant>
- **Agent Skills with Anthropic**
<https://www.deeplearning.ai/courses/agent-skills-with-anthropic>
- **Building toward Computer Use**
<https://www.deeplearning.ai/courses/building-toward-computer-use-with-anthropic>

Path 5: Your own menu

Discuss your own proposal with the lecturers: you should bring a new proposal for a selection of courses of your own choice that is comparable to the paths above and get approval from one of the lecturers.

Project literature

Each student screens at least ten abstracts and reads at least five papers in depth, and the literature work should support the relevant projects. Where a project provides more papers, students should use that higher number. The team coordinates selections to avoid

unnecessary duplication. Students may use NotebookLM and other AI tools to create supporting learning materials and may include the notebook/interaction URL in the portfolio.

Evidence

For the selected online-learning route, the student includes evidence of completion , student-authored work such as adapted examples, learning highlights, application within the project, and a short reflection. No official certificate from the online-learning provider is required, although students may obtain one if they wish, potentially at an additional cost.

Appendix 3:

Research paper assignment

During the first four weeks of the module, students write a research paper on a topic related to Applied AI. The purpose of the assignment is to develop students' ability to investigate an Applied AI topic in a structured and evidence-based manner and to communicate their findings clearly in an academic paper.

The paper is written in pairs. During week 1, students select a writing partner and develop an initial proposal for the paper. This proposal must include:

- the topic;
- a provisional title;
- a main research question;
- three sub-questions.

Before students start writing the paper, the proposed title, main research question and sub-questions must be discussed with and approved by a lecturer. Students are expected to select a topic that is sufficiently focused, relevant to Applied AI and suitable for investigation within the available time.

The paper should demonstrate appropriate use of relevant and reliable sources. Students are expected to develop a coherent argument, answer the research questions on the basis of the evidence collected, and clearly distinguish between source-based information, their own analysis and their conclusions.

A peer review takes place during week 3. Each paper is assigned to another student for review. An overview will be provided indicating who is responsible for reviewing which paper.

Each pair is responsible for ensuring that the assigned reviewer receives access to the paper in sufficient time. The reviewer uses the paper assessment form provided in Teams and provides substantive feedback, including both strengths and suggestions for improvement. The peer review must be completed before the start of the first scheduled Applied AI session in week 3.

Students are expected to take the peer review very seriously. Useful feedback should be specific, constructive and related to the requirements of the assignment. The authors are expected to consider the

feedback received and, where appropriate, use it to improve the final version of the paper. Evidence of both the feedback received and the feedback provided must be included in the portfolio.

The final paper must be submitted on the Friday of week 4 in accordance with the submission instructions provided in Teams. The paper is assessed as Sufficient or Insufficient using the prescribed paper assessment form. The assessment form and any further requirements concerning structure, length, referencing and submission are available in these appendices and/or the Applied AI Teams environment.

Successful completion of the research paper is a mandatory component of the period 1 Go/no-go assessment. A paper assessed as Insufficient must therefore be improved in accordance with the applicable resit procedure.

Use of the Wednesday sessions

Students are expected to use the Wednesday sessions as the primary opportunity to discuss the setup and development of their paper with the teaching team. Questions concerning the topic, title, main research question, sub-questions, structure, expectations or assessment should therefore be raised during these sessions wherever possible. The Wednesday sessions are specifically intended for consultation, feedback and clarification. Students should make active use of this opportunity before contacting lecturers separately by email or Teams.

Appendix 4:

Working paper

By the end of period 2, each project group is expected to submit a working paper based on the Applied AI project they have carried out during the semester. The working paper is a group deliverable and forms part of the portfolio.

The purpose of the working paper is to present relevant aspects of the project in an academic and evidence-based manner. In the academic field, working papers are used to document and communicate research, analysis, methods, arguments and preliminary or developing findings before they are necessarily published in a final journal or conference format. They allow authors to structure their reasoning, relate their work to existing literature and make their approach and conclusions open to discussion and review.

Within Applied AI, the working paper should therefore demonstrate appropriate use of literature, analysis, argumentation, methodology and reporting, and should relate clearly to the work undertaken within the project.

Before starting the paper, the project group must obtain approval from the supervising lecturer for:

- the proposed title;
- the main research question;
- the sub-questions.

The group is expected to divide the work in such a way that each student makes a clear and substantive contribution to the paper. Each student must be able to identify and explain their own contribution. This individual contribution may include, for example, literature research, methodology, analysis, technical content, interpretation of results, discussion or conclusions.

The working paper is assessed as a group deliverable using the prescribed assessment form. The assessment form and further instructions for the working paper are available in these appendices and/or in the Applied AI Teams environment.

The student's identifiable individual contribution to the working paper will also be used as evidence when assessing their individual contribution to the project.

Use of the Wednesday sessions

Students are expected to use the Wednesday sessions as the primary opportunity to discuss the setup and development of their paper with the teaching team. Questions concerning the topic, title, main research question, sub-questions, structure, expectations or assessment should therefore be raised during these sessions wherever possible. The Wednesday sessions are specifically intended for consultation, feedback and clarification. Students should make active use of this opportunity before contacting lecturers separately by email.

Appendix 5:

Professional skills portfolio

Professional skills and conduct form an important part of the Applied AI module. Students are expected to demonstrate that they can act, communicate and make decisions as responsible IT professionals and that they can critically consider ethical, societal and professional issues related to Artificial Intelligence.

The [ACM Code of Ethics and Professional Conduct](#) provides the main professional framework for this part of the module. Students are expected to use relevant principles from the Code to examine concrete professional situations and to consider the responsibilities associated with the development and application of AI.

The professional skills section of the portfolio consists of two components:

1. Professional Conduct and Ethics Reflection;
2. Ethics and AI Conference.

Claims and reflections should, where reasonably possible, be supported by concrete evidence. Evidence should be selected for its relevance and quality rather than its quantity.

7.1.1 Professional Conduct and Ethics Reflection

Students must select and analyse at least four significant professional situations encountered during the semester. The purpose is to demonstrate how the student has acted and made decisions as an IT professional and how relevant principles from the ACM Code of Ethics and Professional Conduct apply to professional practice.

The situations should together demonstrate different aspects of professional conduct and judgement. They may arise from collaboration within the project team, interaction with the client or other stakeholders, technical or project decisions, the use of AI and data, feedback, disagreements, uncertainties or other situations involving professional responsibility.

At least one of the selected situations must relate directly to an ethical, societal or professional issue arising from the development or use of Artificial Intelligence.

Students should select situations that genuinely required them to consider how they should act as an IT professional. The situations should therefore go beyond simply describing tasks that were completed.

For each situation, the student addresses the following:

- **Situation** - What happened? Briefly describe the professional context.
- **Responsibility** - What was your role and what were you responsible for?
- **Action** - What did you (not) do, or what decision did you (not) make?
- **ACM principle(s)** - Which principle or principles from the ACM Code of Ethics and Professional Conduct were relevant?
- **Stakeholders and considerations** - Who could be affected by the situation or decision? Which interests, risks, responsibilities or possible consequences did you consider?
- **Alternatives** - What other courses of action were available?
- **Justification** - Why did you decide to act in the way you did?
- **Outcome** - What happened as a result?
- **Reflection** - Looking back, do you still consider this to have been an appropriate professional response? What did you learn, and what might you do differently in a similar situation?
- **Evidence** - Where can supporting evidence be found in the portfolio?

Students are not required to address every principle of the ACM Code. They should select those principles that are genuinely relevant to the situations discussed. The purpose is to demonstrate thoughtful application of the Code rather than to complete a checklist.

7.1.2 Ethics and AI Conference

During the second period of the Applied AI module, students contribute to an Applied AI Conference for students from earlier years of the Information Technology and Computer Science programmes. The conference provides an opportunity to share knowledge gained during the module and to practise explaining technical, ethical and professional topics to an audience with less advanced knowledge.

The conference consists of two parts.

The morning programme focuses on Ethics and AI. Applied AI students prepare and deliver lectures, workshops or other suitable educational activities addressing ethical, societal and professional issues related to Artificial Intelligence. Relevant principles from the ACM Code of Ethics and Professional Conduct should be incorporated where appropriate.

Students are expected to investigate their chosen topic in sufficient depth to be able to explain it clearly, discuss relevant stakeholders, risks and consequences, and encourage the audience to think critically about the responsibilities associated with the development and use of AI.

The afternoon programme focuses on the Applied AI projects. Each project group presents its project through a poster presentation and contributes workshops, demonstrations or other educational activities that introduce students from earlier years to technologies, techniques, methods or approaches used within the project.

The emphasis is not solely on presenting what the project has produced. Students should use the project as a basis for teaching others about relevant technical concepts and professional approaches. Every member of the project group must make an active and identifiable contribution to the conference.

Students are expected to adapt their communication to the knowledge level of the audience, explain complex topics clearly and respond professionally to questions, discussion and different viewpoints.

The organisation and practical coordination of the conference are the responsibility of the teaching team and support staff. Students are responsible for the preparation, substantive quality and delivery of their own contributions.

The portfolio must contain evidence of the student's individual contribution to the conference. This should include:

- the student's role in preparing and delivering the conference activities;
- the topic or topics addressed and their relevance;
- for the morning programme, the main ethical, societal or professional issues considered and, where relevant, their relationship to the ACM Code of Ethics and Professional Conduct;
- for the afternoon programme, the technologies, techniques, methods or approaches explained to the audience;
- the intended learning goals for the audience;
- materials developed for lectures, workshops, poster presentations, demonstrations or other activities;
- evidence that the student actively contributed to the delivery of the activities;

- relevant questions, discussions, feedback or responses from participants;
- a concise reflection on what the student learned from preparing, explaining and discussing the topics with others.

The conference provides evidence for the assessment of professional skills and conduct, particularly professional communication, knowledge sharing, responsibility, ethical awareness and the ability to communicate with different audiences. Students are not assessed on the practical organisation or logistics of the conference.

7.1.3 Evidence and assessment

The evidence submitted under this section forms part of the assessment of professional skills and conduct and will also be discussed during the individual criterion-based interview. Students must therefore be able to explain and justify the situations, decisions, ethical considerations and conference contribution included in their portfolio.

A broader reflection on the student's development as an IT professional, including explicit reflection on the ACM Code of Ethics and Professional Conduct, is included in the overall reflection elsewhere in the portfolio.

The requirements in this appendix describe what students must demonstrate and include in their portfolio. The quality and level of achievement are assessed using the Professional Skills Scoring Rubric included separately in the appendices.

Appendix 6:

External client feedback

As part of the final assessment, each student must obtain feedback from the external client involved in the Applied AI project.

The purpose of this feedback is to provide the examiners with an external professional perspective on the student's performance during the project. The client is in a particularly good position to comment on aspects such as the student's professional behaviour, communication, independence, ability to apply knowledge, professional judgement and contribution to the project.

The feedback form is based on the approach used during the graduation phase and includes elements derived from the Dublin Descriptors. The form enables the client to provide feedback on different aspects of the student's performance and to add explanatory comments.

The Dublin Descriptors describe broad capabilities expected at Bachelor level, including knowledge and understanding, application of knowledge, formation of judgement, communication and learning skills. They are used here as a framework for client feedback, not as a separate assessment by the client.

Each student is responsible for:

- providing the feedback form to the external client;
- ensuring that the client has sufficient opportunity to complete the form;
- collecting the completed form before the portfolio submission deadline;
- including the completed feedback form in the final portfolio.

Students should therefore arrange the feedback in good time and should not wait until shortly before the final portfolio deadline.

The external client feedback is important because it provides evidence from someone who has observed the student functioning within the professional project environment. It complements the evidence provided by the student and helps the examiners form a well-supported judgement of the student's individual performance.

The client feedback is advisory. The external client does not determine the student's grade. The examiners remain responsible for the final assessment and consider the client feedback together with the other evidence contained in the portfolio.

The feedback form itself is not included in this module book. It will be made available to students in the Teams environment during the second period of the Applied AI semester.

Appendix 7:

Criterion-based interview

The criterion-based interview forms a separate component of the final assessment of Applied AI and is assessed as Sufficient or Insufficient.

The portfolio is the primary evidence base for the final assessment. The purpose of the criterion-based interview is to verify that the evidence presented in the portfolio genuinely reflects the student's own work, understanding and level of achievement. It also gives the student an opportunity to explain, clarify and justify important aspects of the evidence submitted.

The interview does not replace the portfolio. Evidence that should reasonably have been included in the portfolio cannot be introduced for the first time during the interview simply because it was missing from the submission. The interview is used to examine and discuss the evidence that the assessors have already reviewed.

The interview does not introduce new learning outcomes, competences or substantive assessment requirements. Questions are based on the portfolio, the learning outcomes, the HBO-i competences and the assessment components described elsewhere in this module book. The interview is recorded in accordance with the applicable assessment procedures.

What is a criterion-based interview?

A criterion-based interview is a structured professional discussion between the student and the assessors. Rather than following a fixed list of questions, the assessors select questions that help them establish whether the student understands, owns and can justify the work presented in the portfolio.

Students should therefore be prepared to locate relevant evidence in their portfolio, explain what they personally contributed, justify important choices and discuss what they learned from the work. Follow-up questions may be asked when the assessors need further clarification or want to explore the student's reasoning, understanding or reflection in greater depth.

Preparation and conduct

Before the interview, the assessors review the portfolio and identify areas they wish to discuss.

During the interview:

- the student should have the portfolio and relevant supporting evidence readily available;
- the assessors may ask the student to locate and explain specific evidence;
- questions may concern any relevant part of the portfolio;
- students are expected to distinguish clearly between their own contribution and work carried out by the project group;
- students should be able to explain and justify their choices rather than merely describe what was done;
- assessors may ask follow-up questions concerning alternatives, limitations, feedback, consequences or lessons learned.

The interview is assessed as Sufficient when the student demonstrates adequate ownership, understanding and professional reasoning across the relevant parts of the portfolio. It is assessed as Insufficient when the student is unable to adequately explain, justify or demonstrate ownership of substantial parts of the evidence on which their achievement is based.

Possible discussion areas and example questions

The questions below are examples only. They are not a checklist, and students should not expect all questions to be asked. The exact questions will depend on the student's portfolio, project, learning path and the areas the assessors need to clarify or verify.

1. Individual contribution to the project

Possible questions include:

- What did you personally contribute to the project, and where can this be seen in the portfolio?
- Which part of your contribution do you consider most substantial, and why?
- Why did you choose this approach rather than an alternative?
- How did you test, evaluate or otherwise assess the quality of your work?
- What changed as a result of feedback, experimentation or an earlier iteration?
- What are the most important limitations of your contribution?
- How does your work relate to the wider project?
- What did you personally contribute to the project group's working paper?
- Which technical learning outcomes or HBO-i competences are demonstrated by this work?

Relevant evidence may include analyses, designs, experiments, code, AI components, tests, evaluations, Scrum artefacts, feedback,

repository history, contributions to the working paper and other project evidence.

2. Learning path

Possible questions include:

- Which project-related knowledge or skills did you need to develop?
- Why did you select this particular learning route or resource?
- What did you actually learn?
- Where did you apply this knowledge or skill within the project?
- How did the literature you studied influence your understanding or project work?
- Did your learning needs change as the project developed?
- What would you need to learn next if the project continued?

Students should be able to show evidence of their project-aligned learning route, student-authored learning activities, application within the project and relevant literature study.

3. Professional skills and conduct

Possible questions include:

- Which professional situation best demonstrates how you acted as an IT professional?
- Which stakeholders and responsibilities were involved?
- Which principles from the ACM Code of Ethics and Professional Conduct were relevant?
- Which alternatives, risks or possible consequences did you consider?
- Looking back, would you make the same decision again?
- What did you learn from the situation?
- What was your contribution to the conference?
- How did you adapt your explanation or activity to students from earlier years?
- What did you learn from presenting ethical issues, technologies or techniques to others?

Relevant evidence may include the Professional Conduct and Ethics Reflection, project situations, conference materials, poster presentations, workshops, feedback and other evidence of professional conduct.

4. Learning outcomes and HBO-i competences

Possible questions include:

- Which evidence best demonstrates this learning outcome?
- Why does this evidence demonstrate achievement rather than merely participation?
- What was your individual contribution to this evidence?

- Why do you believe this work demonstrates the required HBO-i competence level?
- What aspects of complexity, independence, responsibility or professional quality can be seen in the evidence?
- Which evidence would you consider strongest, and why?

Students should be able to explain the relationship between their evidence, the learning outcomes and the relevant HBO-i competences.

5. Overall reflection and development

Possible questions include:

- How have you developed during the Applied AI module?
- Which experience had the greatest influence on your development?
- What are your main strengths at the end of the module?
- Where do you still see limitations or areas for further development?
- Which feedback had the greatest impact on your work or professional development?
- What would you approach differently if you were starting the project again?
- How has your understanding of Applied AI and professional IT practice changed during the semester?

The assessors may also ask the student to locate and explain specific evidence referenced in the overall reflection.

Important note

The questions in this appendix are examples intended to help students understand the nature of the criterion-based interview. They do not constitute a fixed question set, and there is no requirement for assessors to ask every question or to cover every discussion area in the same way for every student.

The interview is tailored to the student's portfolio and is intended to establish whether the student can demonstrate ownership, understanding and professional reasoning in relation to the evidence submitted.

Appendix 8:

Grade calculation

The detailed assessment form used by the examiners is available in the Teams environment and is not reproduced in this module book. This appendix explains how the assessment of Applied AI is structured and how the final result is determined.

The assessment consists of three components:

1. Go/no-go assessment
2. Portfolio assessment
3. Criterion-based interview

All three components must be completed successfully in order to pass the module.

1. Go/no-go assessment

The Go/no-go assessment takes place at the end of period 1 and is assessed as Sufficient or Insufficient. The assessment is based on three mandatory elements:

- the period 1 research paper;
- successful completion of the prescribed Path 0 online learning;
- sufficient individual contribution to the project during period 1.

All three elements must be sufficient in order to receive a Go. The elements cannot compensate for one another. A Go is required to continue with the module in the normal manner. The Go/no-go assessment does not contribute numerically to the final grade.

2. Portfolio assessment

The final portfolio is assessed numerically. The portfolio assessment consists of six weighted components:

- Individual contribution to the project – 35%
- Learning path – 25%
- Professional skills and conduct – 15%
- Learning outcomes and competence justification – 10%
- Overall reflection – 10%
- Quality and readability of the portfolio – 5%

Together, these components determine the portfolio grade.

The portfolio is assessed entirely on an individual basis. Although students work in project groups, the project itself and the overall project result are not graded. The assessment concerns the individual student's contribution, learning, professional development, evidence and demonstrated level.

External client feedback forms part of the evidence considered by the examiners. The feedback is advisory and does not independently determine the student's grade.

The detailed criteria and guidance used for the portfolio assessment are included in the assessment form available in Teams.

3. Criterion-based interview

The criterion-based interview takes place as part of the final assessment and is assessed as Sufficient or Insufficient. The purpose of the interview is to verify the student's ownership, understanding and professional reasoning in relation to the evidence presented in the portfolio. Students must be able to explain and justify their individual contribution, learning, professional choices, learning outcomes and HBO-i competences. The interview does not replace missing portfolio evidence and does not contribute numerically to the portfolio grade. Further information about the criterion-based interview, including example discussion areas and questions, is provided in the relevant appendix.

Final result

To successfully complete Applied AI, the student must:

- receive a go (Sufficient) for the go/no-go assessment;
- obtain a portfolio grade of at least 5.5;
- receive a Sufficient result for the criterion-based interview.

Failure to meet any one of these requirements means that the module has not yet been completed successfully. The assessment form used by the examiners, including the detailed portfolio components and weighting, is available in the Teams environment.

